

Adroit Technologies

Protocol Driver

Name	Cimbria Serial
DLL	CIMBRIA



COPYRIGHT

Copyright © 2022 Advanced Worx 112 (Pty) Ltd. All rights reserved.

Information in this document is subject to change without notice. The software described in this document is furnished under a license agreement or nondisclosure agreement. The software may be used or copied only in accordance with the terms of those agreements. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or any means electronic or mechanical, including photocopying and recording for any purpose other than the purchaser's personal use without the written permission of Advanced Worx 112 (Pty) Ltd.

Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa

<https://adroittech.co.za/>

Contents

1. Overview	4
2. Revisions.....	10
2.1. Document Revision.....	10
2.2. DLL Revision	10
3. Wiring	5
4. Device Configuration	6
4.1. Device Configuration	6
4.1.1. Configuration Selection [not implemented].....	6
4.1.2. Port Setup.....	6
4.1.3. Device Details.....	7
5. Supported Addresses	8
6. Protocol Definition and Considerations	9
6.1.1. Cimbria Protocol Instruction Structure.....	9
7. General Notes and Troubleshooting	10
7.1. Contact Details.....	10
7.2. Adroit Technologies Knowledge Base.....	10

1. Overview

This document describes the setup of, and the communication protocol implemented in the Adroit protocol driver, with the following logic unit: Cimbria level sensor.

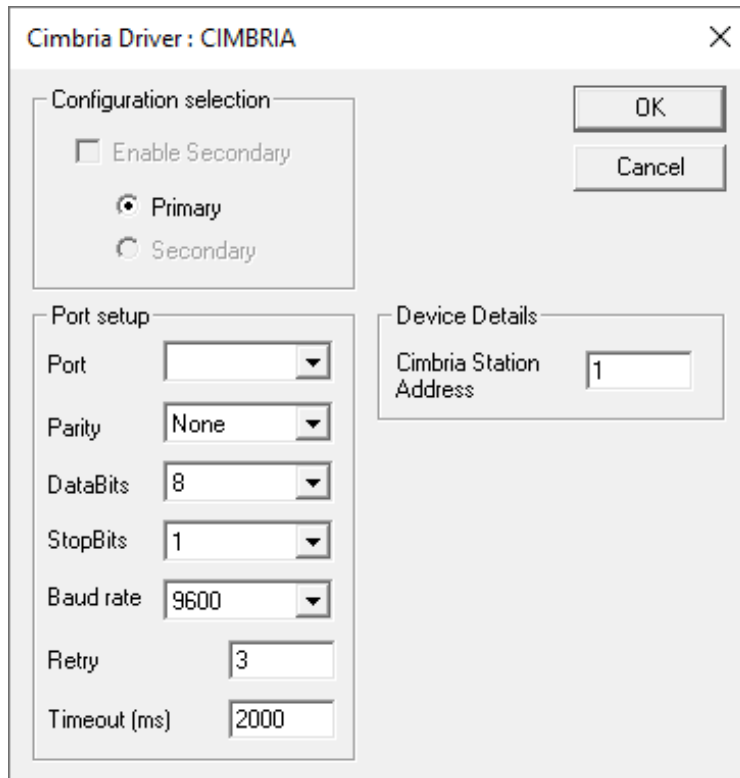
The Cimbria devices can take up to 12 seconds to reply to a single line request, for this reason it is recommended that scans are only added when the device is stopped.

2. Wiring

Standard serial interconnection (RS-485).

3. Device Configuration

3.1. Device Configuration



3.1.1. Configuration Selection [not implemented]

Enable Secondary – Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary/Secondary – Select the primary or secondary button depending on which connection you wish to configure.

This feature is not available for this driver.

3.1.2. Port Setup

Port – Select the serial port to use for communications.

Parity – Select even or uneven parity in a single serial transmission message.

DataBits – Select the number of data-bits per packet.

StopBits – Select the number of stop-bits per packet.

Baud Rate – The transmission rate of data (including start and stop bits). The default baud rate for a Cimbria unit is 9600.

Retry – The number of times to retry a transaction before a comms-failure is reported.

Timeout – The number of milliseconds to wait between communication-retries.

3.1.3. Device Details

Cimbria Station Address – Supply the unit address of the Cimbria device to communicate with.

4. Supported Addresses

The Adroit Cimbria serial driver supports the following data address types:

Description: Sensor average value request

Address Format: <Instruction Code> <Line Number>:<Sensor Number>

- **Instruction Code:** A
- **Line Number:** 0 – 8
- **Sensor Number:** 1- 24

Adroit Data types Supported: Integer, Real

Example Address: "A 0:1", "A 8:24"

Description: Sensor temperature value request

Address Format: <Instruction Code> <Line Number>:<Sensor Number>

- **Instruction Code:** T
- **Line Number:** 0 – 8
- **Sensor Number:** 1- 24

Adroit Data types Supported: Integer, Real

Example Address: "T 0:1", "T 8:24"

Description: Last measured sensor offset request

Address Format: <Instruction Code> <Index>

- **Instruction Code:** F
- **Data Request:** 1 – 2 (1 = offset value, 2 = reference value)

Adroit Data types Supported: Integer, Real

Example Address: "F 1", "F 2"

Description: Port (A, B, C) state request

Address Format: <Instruction Code> <Index>

- **Instruction Code:** P
- **Data Request:** 1 – 3 (1 = Port A, 2 = Port B, 3= Port C)

Adroit Data types Supported: Integer, Real

Example Address: "P 1", "P 3"

Description: Firmware version request

Address Format: <Instruction Code>

- **Instruction Code:** V

Adroit Data types Supported: String

Example Address: "V"

Description: Current 12V reading request

Address Format: <Instruction Code>

- **Instruction Code:** W

Adroit Data types Supported: Integer, Real

Example Address: "W"

5. Protocol Definition and Considerations

5.1.1. Cimbria Protocol Instruction Structure

Message Start	Cimbria Station Address	Command/ Instruction	Command Parameters	Message End
!	AAA	C	P...	*

Message start:

The exclamation mark character (0x21) indicates the start of a new instruction.

Cimbria Station Address:

This three digit number indicates the address of the Cimbria device, to which the message will to be sent.

Command/Instruction:

This single character instruction identifies the type of information that is requested by the user. This could be any one of the instructions described in the previous section ([Supported Addresses](#)).

Command Parameters:

These parameters are command specific and vary in length between commands.

Message stop:

The Asterix character (0x2A) indicates the end of the instruction.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	18-Jan-2023	K. Robinson	Changes to format, updated image resources. Added Appendix B DCOM Hardening. Removed Windows XP section	D. Jordaan
2.0				
3.0				

7.2. DLL Revision

Rev.	Date	Changes
	24-Nov-2014	First Release notes!
	26-June-2018	Extended address ranges of A and T line numbers to 0 - 8
4	18-Aug-2020	Ongoing driver maintenance.

